

a. & b. GenAI Programs and Access Method

- **GenAI 1:** Google Gemini (Accessed via the Gemini web interface).
- **GenAI 2:** OpenAI ChatGPT (Accessed via the ChatGPT web interface).

c. Prompt Used

"Write a C program that picks a fixed secret number (7) between 1 and 10. Give the user 3 tries to guess it using a loop, printf, and scanf. Use if-else for too high, too low, or correct. Stop the loop if correct and print a win/lose message."

d. Generated Code

GenAI 1 (Gemini) Raw Code:

```
#include <stdio.h>

int main() {

    int secret = 7, guess, attempts = 0;

    printf("Guess a number between 1 and 10.\n");

    while (attempts < 3) {

        attempts++;

        printf("Attempt %d/3. Enter your guess: ", attempts);

        scanf("%d", &guess);

        if (guess == secret) {

            printf("Correct! You win!\n");

            return 0;

        } else if (guess > secret) {

            printf("Too high! Try again.\n");

        } else {

            printf("Too low! Try again.\n");

        }

    }

}
```

```
    printf("You lose! The secret number was %d.\n", secret);  
  
    return 0;  
  
}
```

GenAI 2 (ChatGPT) Raw Code:

```
#include <stdio.h>  
  
int main() {  
  
    int secret = 7, guess;  
  
    printf("Guess a number between 1 and 10.\n");  
  
    for (int i = 1; i <= 3; i++) {  
  
        printf("Attempt %d/3. Enter your guess: ", i);  
  
        scanf("%d", &guess);  
  
        if (guess == secret) {  
  
            printf("Correct! You win!\n");  
  
            break;  
  
        } else if (guess > secret) {  
  
            printf("Too high! Try again.\n");  
  
        } else {  
  
            printf("Too low! Try again.\n");  
  
        }  
  
        if (i == 3) printf("You lose! The number was %d.\n", secret);  
  
    }  
  
    return 0;  
  
}
```

e. Code Analysis

- **Correctness:** Both programs correctly implement the basic logic, loop 3 times, and successfully evaluate the guess.
- **Execution Time:** Both operate in $O(1)$ constant time since the loop runs a maximum of 3 times regardless of input.
- **Space Complexity:** Both operate in $O(1)$ space complexity, allocating only basic integer variables in memory.
- **Maintainability:** The Gemini code used a while loop with a separate attempt counter, making it slightly easier to read and modify than the compact for loop provided by ChatGPT.

f. Selection & Justification

I selected the Gemini (Program 1) code as the baseline. The explicit while loop structure makes it easier to inject edge-case handling (like invalid inputs) without cluttering a for loop header.

g. Code Improvements

To improve *correctness*, I added an edge-case handler to check if the user input is less than 1 or greater than 10. If so, it alerts the user and does not count against their 3 attempts. For *maintainability*, I added extensive line-by-line comments and prologue documentation. The *execution time* remains $O(1)$ constant time despite the added check, and *space complexity* remains $O(1)$ as no new major data structures were added.